

Introduction:-

① Artificial Intelligence and its Applications.

What is AI?

- AI is the branch of computer science that deals with making computers or machines behave in ways that are considered "smart" or "intelligent" like humans.

Main goal of AI:

- To create systems that can think, learn, solve problems; and make decisions like humans.

Applications of AI:

→ Domain	Example
Healthcare	Disease prediction, Robotic surgery
Transport	Self driving cars
Finance	Fraud detection, Algorithmic trading
Education	Personalized tutoring
Retail	Chatbots, product recommendation
Agriculture	Crop health monitoring and using drones.

② Artificial Intelligence Techniques:

AI uses many techniques to perform tasks:

a) Search Algorithms:

- used to find the best solution among many (eg, finding shortest path in a map). (Best way among all)

b) Knowledge Representation

→ method to store facts and rules in a way that a machine can understand (eg logic, semantic networks).

c) Machine learning

→ Enables machines to learn from data without being explicitly programmed

d) Natural Language Processing (NLP)

→ Allows machines to understand and respond to human language

e) Computer vision:

→ Helps machines "see" and understand images or videos.

③ Level of models (Types of AI systems).

① Types of AI Based on Functionality:

i) Reactive machines

- No memory or past experience
- Just responds to current input
- ex: IBM's Deep Blue Chess Program

ii) Limited Memory

- Can learn from past data for a short time.
- ex: self-driving cars analyzing nearby vehicles

iii) Theory of mind

- understands people's emotions, beliefs and thoughts.
- still under development.

iv) Self-aware AI

- Aware of its own existence and emotions
- still a futuristic concept.

③ Types of AI Based on capabilities:-

(Also called Level of Intelligence or Capability Model)

Type	Description	ex
ANI- Artificial Narrow Intelligence	weak AI, does one specific task	Google, Siri
AGI- " General " " " " " " "	Thinks and learn like human	Still a research
ASI- " Super " " " " " " "	more intelligent than human	Future concept

④ Criteria of Success in AI

An AI system is considered successful if it has:

Accuracy:

- The AI system makes correct decisions, predictions, and outputs with minimal errors.

Examples: Medical diagnosis AI, Translation system and Fraud detection.

Efficiency

- The AI performs tasks quickly while using minimal computational resources like memory, processing power and energy.

Examples: Mobile AI assistants, Real-time recommendations and Autonomous vehicles.

Adaptability

- The AI can handle new situations, environments, or data that it hasn't encountered during training.

Example: Chatbots, Game AI, weather prediction.

Usability

- The AI system is intuitive, accessible and easy to use to interact with effectively.

ex: voice interface, photo editing apps, search engines.

⑤ Intelligent Agents:

- An agent is anything that perceives its environment and takes actions to achieve goals.

Types of Agents :-

1) Simple Reflex agents

- Acts only based on current input without considering past events or future consequences.
- ex: Automatic doors, smoke alarm.

② Model-Based Agent:

→ Maintains internal memory of previous states and uses this info to make decisions.

Ex: GPS navigation, Email spam filter, chess AI.

③ Goal-Based Agents:

→ Chooses actions specifically designed to achieve predetermined objectives or goals.

Ex: Delivery robot, stock trading bot, cleaning bot.

④ Utility-Based Agent

→ Evaluates different possible outcomes using a utility function to choose the best possible action.

Ex: Ride-sharing app, Investment advisor AI.

⑤ Learning Agent

→ Improves performance over time by learning from past experiences and feedback.

Ex: Recommendation systems, personal assistance, Game AI.

⑥ Nature of Agents:

An agent has:

i) Sensors

- Input devices that allows the agent to perceive and gather information from its environment.

Ex: Robot vacuum, Autonomous car, smart home.

2) Actuators:

output devices that allow the agent to perform actions and affect its environment

Ex:

Industrial robot, Drone, smart speakers.

3) Percept:

- The specific information or data that an agent receives from its sensors at any given moment.

Ex: Security camera AI, weather monitoring sys, chatbot.

4) Performance Measure

- Metrics used to evaluate how well the agent accomplishes its intended tasks and goals.

Ex: Search engine, medical diagnosis AI, Trading algorithm.

⑦ Learning Agents:

- Learning agents improve automatically through experience by adapting their behaviours based on feedback and new information.

Components:-

1) Learning Element:

- The component that analyzes experiences and feedback to update the agents knowledge and improve future performance.

Ex:- Spam Email, Autonomous vehicles etc.

2) Performance elements

- The components responsible for selecting and executing actions based on the agent's current knowledge and capabilities.
- Ex: Chess AI, Stock trading.

3) Critic

- Evaluates the agent's actions and provides feedback on performance, indicating whether actions were good or bad.
- Ex: Game AI, voice assistant.

4) Problem Generator:

- Deliberately creates new simulations or suggests exploratory actions to help the agent gain broader experience & learning opportunities.
- Ex: online learning platform, Exploring robots.

All works together to give and make a good learning agent.

⑧ Advantages of AI

1) Fast and Efficient:

- AI systems can process vast amounts of data and perform complex calculations much faster than humans, completing tasks in seconds that would take human hours or days.
- Ex: Financial trading, image recognition, language translation.

2) No fatigue - can work 24/7?

→ AI systems don't get tired, need breaks or suffer from decreased performance due to exhaustion, allowing continuous operation without interruption.
ex: factory robots, Autonomous vehicles.

3) Memory-Based (Retains large data accurately)

- AI systems can store, access, and recall massive amounts of information with perfect accuracy, never forgetting details or making memory-related errors.

ex: Medical AI, Legal research AI, Navigation system.

4) Risky Jobs (used in Hazardous environments)

- AI powered robots and systems can operate in dangerous conditions where human presence would be life-threatening or impossible.

ex: space exploration, Deep-sea, Bomb disposal, volcano.

5) Consistency

6) Scalability

7) Precision

⑨ Limitations of AI

1) High cost

- AI systems require significant financial investment for development, implementation, hardware, maintenance, and skilled personnel, making them expensive for many organizations.

ex: Healthcare AI, Data centers,

② No ~~an~~ emotional Intelligence

- AI lacks the ability to understand, interpret, or respond appropriately to human emotions, empathy, and social nuances that are crucial in many interactions.

Ex: Customer service, HR decisions.

③ Lacks creativity:

- AI cannot generate truly original ideas, think outside established patterns or create innovative solutions that require imagination and abstract thinking.

Ex: Art and design, Problem-solving.

④ Dependent on Data and programming

- AI systems can only work within the constraints of their training data and programming, making them inflexible when facing situations outside their designed parameters.

Ex: Biased datasets, Limited scenarios, medical AI.

Lack of common sense

No consciousness

Ethical concerns

Job displacements

⑩ Impact and Examples of AI
AI is transforming every field with revolutionary applications.

I) Security:

- Enhanced Surveillance, threat detection, and crime prevention through intelligent monitoring system
- Ex: Facial recognition, Fraud detection,

II) Education:

- Personalized learning experiences and automated education content creation tailored to individuals student needs.

Ex = AI-based test generator, Language learning

III) Retail:

- Streamlined operations, inventory management, and enhanced customer shopping experiences.

Ex: Automated warehouse, cashier-less stores.

IV) Entertainment:

- More immersive and interactive entertainment experience with intelligent content creation

Ex: content creation, AI in game, etc.

Healthcare

Transportation

Finance

Agriculture.

⑪ Application. Domains of AI.

Robotics

Gaming

Natural language processing

Computer vision

Data Analytics

Medical Diagnosis

Autonomous vehicles

Smart-Homes

Healthcare robots

⑫ The AI Ladder - The Journey to Adopt AI

IBM proposed a ladder to help companies adopt

AI :

- i) Collect : Gather relevant data
- ii) Organize : Clean and label the data
- iii) Analyze : Apply AI/ML to gain insights
- iv) Infuse : Integrate AI into business processes
- v) Modernize : update continuously for improvements.

⑬ Advice for a career in AI:

- Learn Python, Java and libraries like TensorFlow, PyTorch
- Understand math: Linear Algebra, probability & statistics.
- Learn ML & Deep Learning
- Work on real-life projects and datasets
- Participate in competitions like Kaggle, Hackathons

⑭ Hotbeds of AI Innovation:

Countries leading in AI research and startups:

- i) USA - Silicon Valley
↳ Autonomous vehicles, Big tech AI, cloud computing
- ii) China - Beijing, Shenzhen
↳ Facial ~~recognition~~ recognition, Healthcare, smart cities
- iii) India - Bengaluru, Hyderabad
↳ Healthcare, Agriculture, Education
- iv) UK - London
↳ Deepmind, Finance, AI government

Unit
2:Problem Solving in AI (steps)

- Initial state
- Goal state
- Set of actions
- Path cost

1) State space search

- All possible states that can be reached to the goal state from start state with valid actions

Eg: Solving a puzzle game

1 2 3	move down	1 2 3	move right	1 2 3
4 _ 6	→	4 5 6	→	4 5 6
7 5 8		7 _ 8		7 8 _

Note: uninformed state search (cost is not given)

ii) Control strategies

→ How to explore the state space

1) uninformed (Blind) search : No knowledge (only problem)

eg: BFS, DFS

ii) Informed (Heuristic) search : use additional knowledge

eg: Best-First search

Heuristic: 2 or 3 steps

BFS (Queue) : level order FIFO

DFS (stack) : depth node, FILO

Best-First Search : priority queue data structure

Problem characteristics:-

- i) Deterministic / Non-deterministic: Predictable (fixed result) / Non-predictable (no result)
- ii) Static / dynamic: Fixed for someone ^{medical report} / Different for many persons (results)
- iii) Single-agent / multi-agent: one agent involvement / online multiplayer games.
- iv) Discrete / Continuous: Puzzle games / Drone camera not fixed angle.

Production System characteristics. (to solve problem according to rule)

- i) Simplicity: Easy to understand rules and methods. if → then
- ii) Modularity: Easy to add or remove the rule. e.g.: if then
- iii) Data driven: SFS works on input facts.
- iv) Goal oriented: SFS is designed to reach specific goals with fixed rules.
- v) Forward chaining: multiple facts and then rule: suggests step by step to result.
eg: cough and fever → Rule applied

→ Then fever → Then take medicine.

- vi) Backward chaining: starts from a goal and gets facts?
Goal: It is fever? → check if fever and cough are true

- vii) Conflict resolution: where one or more rule then select better one.

Generate and Test:

→ Problem solving method where possible solutions are generated. The solution is tested whether it meets the goals.

Process:-

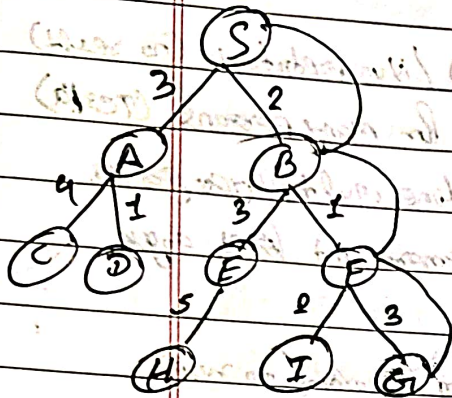
Generate solution

Test solution

Ex: Guessing a password

$$h(n) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Best-First Search.



Node $H(n)$

A 12

B 4

C 7

D 3

E 8

F 2

H 4

I 9

G 13

1) open list: [H, E, I, G]

2) close list: [S, B, F, A]

Hill Climbing Algorithm

→ This algorithm is used to solve those problems which probably have multiple solutions.

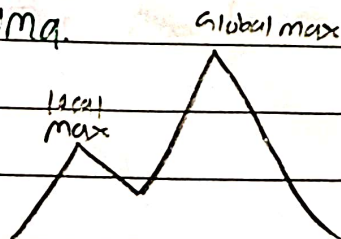
work:

→ If we are trying to hike up the tallest hill, but we can only see the land around us like only forward and backward movement (neighbour path).

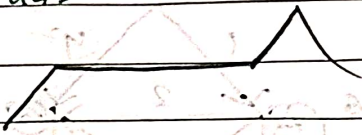
→ The next step must be better than current step.

Limitations:-

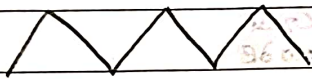
1) Local maxima.



ii) Plateau:



iii) Ridge:



multiple goal state

Solutions for the limitations:-

i) Backtrack few nodes and try new path. (local maxima)

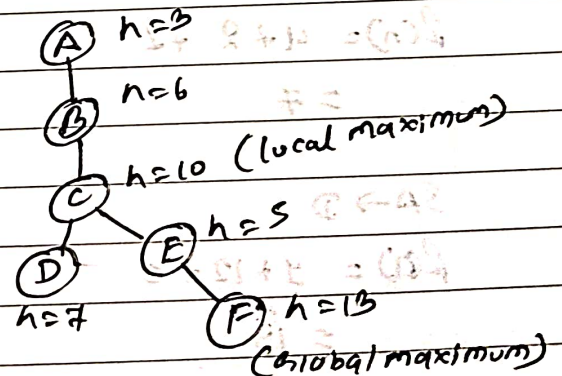
ii) Take a big jump and restart from a new area (Plateau, Ridge)

Important points:-

i) Local search Algorithm

ii) memory efficient (current and prev step only)

iii) used for problems which have multiple solutions

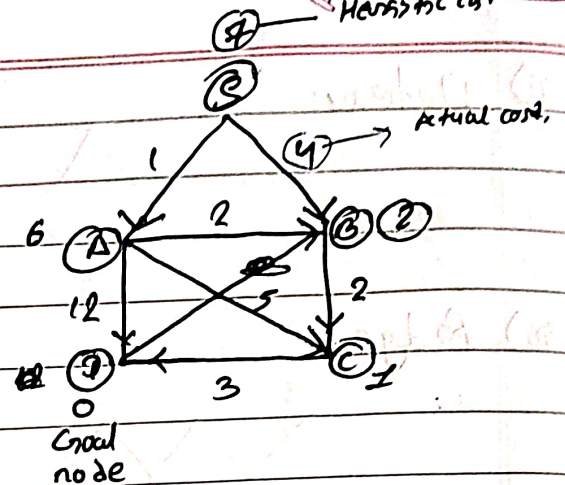


A* Search Algorithm :-

$h(n)$ = Heuristic cost

$g(n)$ = Actual cost between node

$$f(n) = g(n) + h(n)$$



Possible steps

$$S \rightarrow A$$

$$f(n) = g(n) + h(n)$$

$$f(n) = 1 + 6$$

$$\therefore f(n) = 7$$

$$S \rightarrow B$$

$$f(n) = 2 + 4$$

$$= 6$$

$$S \rightarrow C$$

$$S \rightarrow B$$

$$f(n) = 4 + 2 + 1$$

$$= 7$$

$$S \rightarrow A$$

$$f(n) = 1 + 12 + 0$$

$$= 13$$

$$\rightarrow C$$

$$S \rightarrow A \rightarrow B = 1 + 2 + 2 + 1$$

$$= 6$$

$$S \rightarrow A \rightarrow B \rightarrow C = 1 + 2 + 2 + 3 + 0$$

$$= 8$$

$$S \rightarrow A = 7$$

$$S \rightarrow B = 6$$

$$S \rightarrow C = 7$$

$$S \rightarrow A \rightarrow D = 13$$

$$S \rightarrow A \rightarrow B \rightarrow C = 6$$

$$S \rightarrow A \rightarrow B \rightarrow C \rightarrow D = 8$$

$$S \rightarrow B \rightarrow C \rightarrow D = 9$$

$$S \rightarrow B \rightarrow C \rightarrow D = 4 + 2 + 3 + 0$$

$$= 9$$

CONSTRAINT SATISFACTION PROBLEM:

CSP consists of three components V, D, C

- V is a set of variables $\{v_1, v_2, \dots, v_n\}$
- D is a set of domains $\{D_1, D_2, \dots, D_n\}$ one for each variable
- C is a set of constraints that specify allowable combination of values. $C = (\text{Scope}, \text{relation})$ & $\{C_1, C_2, C_3, \dots\}$

where:

Scope: is the set of variables that participate in constraint.

Relation: is that defines the values that variable can take

way of representing:

$$V_1, V_2 \quad C_1 = ((V_1, V_2), (V_1 \neq V_2))$$

$$\begin{array}{cc} A & B \\ (1, 2) & (2, 4) \end{array} \quad C_1 = \{(v_1, v_2), (1, 2), (1, 4), (2, 4), (2, 2)\}$$



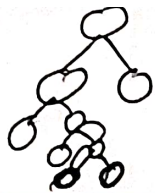
(81) box

(3x3) x 9

variable = space

Domain = (1 to 9) in a box (square)

Constraint = (row, column) no same number



DFS

Time efficient

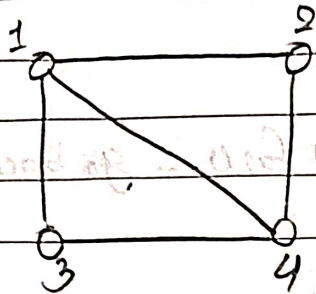
classmate

Date

Page

constraint satisfaction problem (CSP)

Intelligent backtracking = go directed to the conflict point



$V = \{1, 2, 3, 4\}$

$D = \{Red, Green, Blue\}$

$C = \{1 \neq 2, 1 \neq 3, 1 \neq 4, 2 \neq 4, 3 \neq 4\}$

Initial domain

1

2

3

4

RGB

RGB

RGB

RGB

$1 = R$

R

GB

GB

GB

$2 = G$

R

G

GB

B

$3 = G$

R

G

~~GB~~

~~B~~

$3 = G$

R

G

G

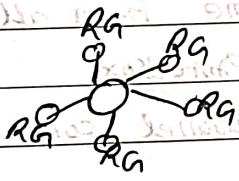
B

Note

if 3 is given 'B' so there is conflict so we backtrack and assign 'G' to 3 and 'B' to 4

Note

Heuristic



without heuristic with heuristic

$2^6 = 32$ ways $\rightarrow 5$ ways

if we fix the central heuristic R then all connected nodes will be G

Techniques of CSP.

- i) Back tracking : exploring different possibilities step by step
- ii) forward checking
- iii) Arc consistency
- iv) minimum remaining value (MRV) / Heuristic

Some techniques of CSP (constraint satisfaction problem)

① Backtracking search

→ The "Try something, see if it works, if it fails -- go back and try something else" method.

Ex: Sudoku Solver - The AI fills one empty cell at a time, checking if it violates the Sudoku rule. If it has a dead end (no valid no), it backtracks and tries a different no.

② Forward checking

→ Whenever you make a choice, you immediately check how it affects future choices, so you don't waste time going down a doomed path.

Ex: Timetable Scheduling → when assigning a class to a time slot, forward checking removes that time from all other courses taught by the same professor.

Also, Assigning tasks to processors in parallel computing.

③ Arc consistency (AC-3)

→ This is about keeping the constraints clean and making sure every variable's choice are still possible given the other choices.

Systematically remove impossible options before doing deeper searches.

Example

Taking family on dinner we just filter out non-veg restaurants at first step.

(4) Minimum Remaining value (MRV)

→ When you don't know where to start, pick the variable with the fewest legal options first. This helps you fail faster if it's impossible, saving time.

Ex: Scheduling a meeting and inviting the most busy person, once their time is fixed then we can manage accordingly.